

Project Report: Smart Sorting Transfer Learning for Identifying Rotten Fruits and Vegetables

1. INTRODUCTION

Project Title:

Smart Sorting Transfer Learning for Identifying Rotten Fruits and Vegetables

Team Members:

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2. PROJECT OVERVIEW

Purpose:

This project aims to reduce food waste and improve quality assurance in agricultural supply chains by using AI to automatically detect rotten fruits and vegetables.

Features:

Image upload feature

AI model for freshness classification

Web-based user interface

Real-time prediction and result display

3. ARCHITECTURE

Frontend:

Developed using HTML, CSS, and JavaScript (served from static folder) for a responsive UI. HTML templates are stored in the templates folder and rendered using Flask.

Backend:

Built with Python using Flask. Handles routing, model inference, and image processing. Core logic resides in `app.py`, and the CNN model logic is encapsulated in `cnn.py`.

Database:

Currently, no persistent database is used. Optionally, a lightweight database like SQLite or MongoDB can be integrated for logging predictions.

4. SETUP INSTRUCTIONS

Prerequisites:

Python 3.x

Flask

TensorFlow/Keras

Installation:

Clone the repository: `git clone https://github.com/your-repo.git`

Navigate to the project directory: `cd your-repo`

Install dependencies: `pip install -r requirements.txt`

Place your model file as `model.h5` in the project root

Ensure folders `media`, `static`, and `templates` are properly populated

Run the app: `python app.py`

5. FOLDER STRUCTURE

`media/` – contains uploaded images

`static/` – contains CSS and JavaScript files

`templates/` – contains HTML files rendered by Flask

`app.py` – main Flask application file

`cnn.py` – defines the model loading and prediction logic

model.h5 – pre-trained CNN model

6. RUNNING THE APPLICATION

Flask Backend (also serves frontend):

Run the following command:

```
python app.py
```

Navigate to <http://127.0.0.1:5000> in your browser to use the app.

7. API DOCUMENTATION

POST /predict

Response:

```
{  
  "status": "success",  
  "prediction": "Fresh"  
}
```

8. AUTHENTICATION

Currently, this project does not use authentication. It is planned as a future enhancement.

9. USER INTERFACE

Upload Image Page

Prediction Result Display

It should be an image with a bar to select images once you select another press predict and you will see results

10. TESTING

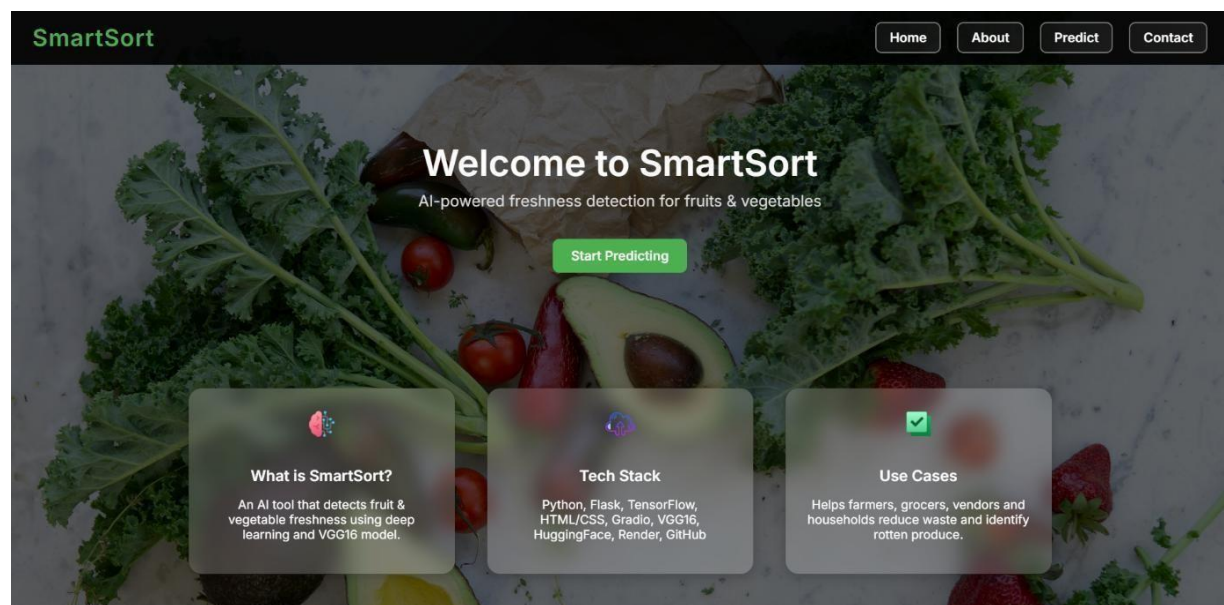
Strategy:

Manual testing of UI interactions

Unit testing using unittest for model functions

Accuracy testing using test datasets

11. SCREENSHOTS OR DEMO



About SmartSort

SmartSort is an AI-powered web application that classifies fruits and vegetables as **fresh or rotten** using deep learning and computer vision.

It is designed to help farmers, food vendors, supermarkets, and consumers quickly assess produce quality and minimize food waste.

With a user-friendly interface and instant image prediction, SmartSort brings the power of machine learning directly to your browser.

Tech Stack

Python

Flask

TensorFlow

YOLOv6

HTML

CSS

JavaScript

Jinja

Render

GitHub

Smart Prediction

Select a fruit or vegetable image to predict its freshness:

[Upload Image](#)

No file selected

[Predict Now](#)

Prediction Result



Apple_Rotten (92.20%)

[Try Another](#)

Result | SmartSort

127.0.0.1:5000/predict

✓ Prediction Result



Banana_Rotten (99.98%)

Try Another

31°C Clear

Search

08:32 PM 06-06-2025

The image shows a web browser window displaying a prediction result from a service called 'SmartSort'. The browser's address bar shows the URL '127.0.0.1:5000/predict'. The main content area features a dark background with a central white square containing a photo of a bunch of bananas. Above the photo, the text '✓ Prediction Result' is displayed in green. Below the photo, the prediction is shown as 'Banana_Rotten (99.98%)'. A blue button labeled 'Try Another' is positioned below the prediction text. The browser window is set against a desktop background showing various fruits and vegetables. The Windows taskbar at the bottom includes a search bar, several application icons, and a system tray showing the temperature as 31°C, the time as 08:32 PM, and the date as 06-06-2025.

12. KNOWN ISSUES

- Image quality affects prediction accuracy
- Limited to trained categories (only trained fruits/vegetables)
- No user login system yet

13. FUTURE ENHANCEMENTS

- Add user authentication system
 - Enable drag-and-drop upload
 - Train model on more fruits/vegetables
 - Develop mobile app version
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